

Loops

Classwork: for, while & Making the Computer Repeat

Name:

Date:

Today we move into VS Code — the real editor developers use — and learn how to make the computer repeat things for us. Always *PREDICT* the output before you run!

Before You Start — VS Code Setup (about 6 min)

Tick these off. Ask for help if you get stuck — then help your neighbour!

1. Open VS Code. Make a new folder for today, then a new file called loops.ts
2. Install the Code Runner extension (Extensions sidebar, search 'Code Runner').
3. IMPORTANT one-time setting so Run works with TypeScript:

Open Settings (Ctrl/Cmd + ,), search for code-runner: executor map, click 'Edit in settings.json', and make sure it has this line:

Add this setting:

```
"code-runner.executorMap": {  
  "typescript": "node $fullFileName"  
}
```

4. Test it! Type the line below, then right-click and choose Run Code.

Type this code:

```
console.log("VS Code works!")
```

Did you see the message in the panel at the bottom? (yes / no)

TIP

Red squiggles still work here! VS Code checks your types as you type, just like the playground did. Run Code is what actually runs the program.

Mission A — Your First Loop (about 8 min)

First, feel the pain. Then fix it with a loop.

Type this (the slow way):

```
console.log(1)  
console.log(2)  
console.log(3)  
console.log(4)  
console.log(5)
```

Now imagine doing this for 1 to 100. How many lines would that be?

Now delete all that and replace it with a loop:

Type this code:

```
for (let i = 1; i <= 5; i++) {  
  console.log(i)  
}
```

Did it print the same thing? What did you see?

Change the 5 to 20 and run again. How many lines did you have to edit?

Mission B — Trace It (about 10 min)

Fill in this table BEFORE you run the code. Then run it and check yourself.

Trace this:

```
for (let i = 1; i <= 4; i++) {  
  console.log(i * 10)  
}
```

Round 1 — i is _____, prints _____

Round 2 — i is _____, prints _____

Round 3 — i is _____, prints _____

Round 4 — i is _____, prints _____

Round 5 — what stops the loop?

Now run it. Were your predictions right?

Mission C — Count Your Way (about 10 min)

Change the three parts of the loop to count differently.

Countdown:

```
for (let i = 5; i >= 1; i--) {  
  console.log(i)  
}  
console.log("Blast off!")
```

What order did the numbers print in?

Skip counting:

```
for (let i = 0; i <= 20; i = i + 5) {  
  console.log(i)  
}
```

What numbers printed?

Your turn: write a loop that counts from 10 up to 100 by tens.

Write your loop here:

Mission D — Loops + Template Literals (about 8 min)

Make your output say something, not just numbers.

Type this code:

```
for (let i = 1; i <= 3; i++) {  
  console.log(`This is round ${i}`)  
}
```

What three lines printed?

Now make a times table! Pick a number and print its table from 1 to 12.

Try this, then change the number:

```
const num: number = 7  
for (let i = 1; i <= 12; i++) {  
  console.log(`${num} x ${i} = ${num * i}`)  
}
```

Which number's table did you make?

Mission E — Add It Up (about 10 min)

An accumulator is a variable you keep adding to. Make it *OUTSIDE* the loop!

Type this code:

```
let total: number = 0
for (let i = 1; i <= 10; i++) {
  total = total + i
}
console.log(`The total is ${total}`)
```

What was the total?

1. Now BREAK it on purpose: move the let total line *INSIDE* the loop and run again.

What happened, and why do you think that is?

CONCEPT

If you build the variable inside the loop, it gets reset to 0 every single round — so it can never add up!

Mission F — Loop + if (about 10 min)

Put a decision inside a loop. Remember % from last week?

Type this code:

```
for (let i = 1; i <= 10; i++) {
  if (i % 2 === 0) {
    console.log(`${i} is even`)
  } else {
    console.log(`${i} is odd`)
  }
}
```

Did every number get labelled correctly?

Challenge: change it so it only prints the *EVEN* numbers (no else needed!).

Write your code here:

Mission G — while Loops & the Infinite Trap (about 10 min)

A while loop repeats as long as something stays true.

Type this code:

```
let lives: number = 3
while (lives > 0) {
  console.log(`Lives: ${lives}`)
  lives = lives - 1
}
console.log("Game over!")
```

What printed?

READ THIS FIRST

DANGER ZONE: next you'll make a loop that never stops. To escape, click the output panel and press Ctrl + C. Your teacher will show you first!

1. Delete the `lives = lives - 1` line and run it. Watch it go forever, then press Ctrl + C.

What happened when you removed that line? Why?

Mission H — Quick Quests (about 10 min)

Small challenges. Check the box as you finish each one!

- ☐ Print your name 5 times using a loop.
- ☐ Count from 1 to 15, but only print the odd numbers.
- ☐ Add up all the numbers from 1 to 100 with an accumulator.
- ☐ Print a countdown from 10 to 1, then 'Happy New Year!'
- ☐ Make a loop that prints the 9 times table.
- ☐ Use a while loop to count down from 5.
- ☐ Print 'Round 1' through 'Round 8' using a template literal.
- ☐ STRETCH: print only the numbers from 1-50 that divide evenly by 7.

Which quest was the trickiest?

Mission I — Mini Project (about 8 min)

Build your own looping program! It must use a loop AND something from a past week (a type, a template literal, or an if).

What does your program do?

Write your code here:

What did it print?

Wrap-Up — Reflect (about 4 min)

What was the hardest part about loops today?

Name something boring a loop could do for you in real life.